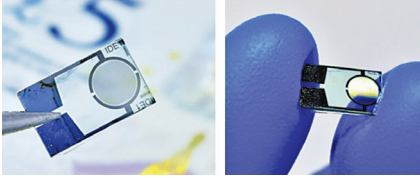


Hydrogen Sensor Enhances Energy Safety

Scientists at the University of Manchester, in collaboration with KAUST, have developed a cutting-edge hydrogen sensor that is compact, affordable, and highly efficient.

Published in *Nature Electronics*, the sensor detects trace



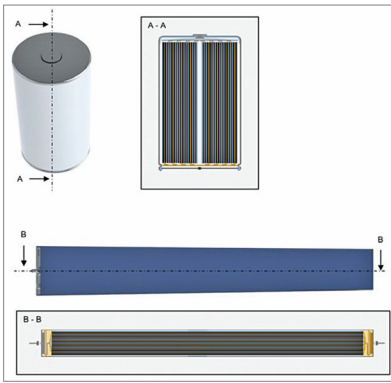
Hydrogen sensor could boost clean energy safety (Credit: www.manchester.ac.uk)

hydrogen within seconds, outperforming commercial detectors. It uses 'p-doping,' where hydrogen reverses oxygen-induced charge

accumulation, causing a rapid, reversible drop in electrical current. Functioning up to 120°C, it detects leaks in pipelines, enclosed spaces, and airborne scenarios via drones. Its ultra-thin, flexible design enables smart device integration, enhancing hydrogen safety and reliability across industries. Researchers aim to improve stability further.

Different Battery Design Approaches

Researchers at RWTH Aachen University analysed Tesla's 4680 Cell and BYD's Blade Cell to compare EV battery technology. Tesla's battery emphasises high energy density, while BYD's



Graphic rendition of the 4680 Tesla Cell (silver cell in the picture) and BYD Blade Cell (blue cell in the picture), including cross sections of both cells (Credit: Jonas Gorsch)

prioritises space efficiency and cost. Cell Reports Physical Science Findings show BYD's battery has better thermal management. The study examined mechanical structure, electrode materials, and thermal performance. Notably, BYD's cell uses a laminated separator, while

Tesla's features a unique binder. Both employ laser welding. Despite differences, their passive component proportions are similar. Further research is needed to assess battery longevity and performance impacts.

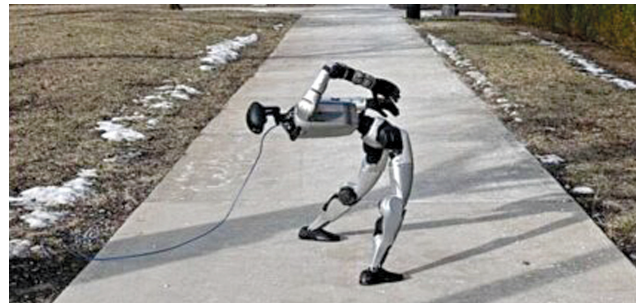
AI Tool For Forensic TBI Investigations

Oxford researchers have developed an AI-driven forensic tool to help police assess traumatic brain injuries with precision.

In collaboration with Thames Valley Police and the National Crime Agency, the team designed a physics-based machine learning framework to predict injury outcomes from forensic data. Trained on 53 anonymised reports, the model achieved 94% accuracy for skull fractures, 79% for loss of consciousness, and 79% for intracranial hemorrhage. Integrating forensic data and computational modeling aids investigations without replacing experts and improves forensic interpretations and legal evidence in assault cases.

Humanoid Robots Learning To Rise After A Fall

Researchers at the University of Illinois Urbana-Champaign have developed HUMANUP, a machine-learning framework enabling humanoid robots to recover from falls autonomously. Using reinforcement learning, HUMANUP operates



Humanoid robot learning to get up (Credit: Xialin He et al)

in two phases: first, identifying effective limb trajectories for standing up, and second, refining motions for smooth, controlled recovery. Addressing challenges like collision geometry and sparse rewards, the framework was tested on the Unitree G1 robot in simulations and real-world settings. Results showed successful autonomous recovery across various terrains, advancing humanoid autonomy and facilitating broader real-world deployment.

Low-Cost Catalyst Enhances Batteries

Researchers have developed a high-performance seawater battery catalyst by combining lignin, a wood by-product (15-35% of wood), with urea from industrial wastewater. By heating lignin and urea to 800°C, nitrogen was incorporated into the lignin structure, enhancing electrochemical reactions and reducing overvoltage. This nitrogen doping lowered energy loss, achieving a power density of 15.76mW/cm²—comparable to platinum (16.15mW/cm²). The catalyst, made from sustainable materials, offers a cost-effective alternative to precious metals, enabling efficient energy storage for seawater batteries and metal-air systems.